

DATE

MONTH

YEAR



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Ass 01

Bubble Sorting

→ Algorithm:

```

for (i = 0, i < n-1; i++)
{
    for (j = 0, j < n-1-i; j++)
    {
        if (A[j] > A[j+1])
        {
            int temp = A[j];
            A[j] = A[j+1];
            A[j+1] = temp;
        }
    }
}

```

→ Example:

14	15	5	7	4
----	----	---	---	---

①

14	15	5	7	4
----	----	---	---	---

14	5	15	7	4
----	---	----	---	---

Date

Month

Year



14	5	7	15	4
----	---	---	----	---

14	5	7	4	15
----	---	---	---	----

(2)

14	5	7	4	15
----	---	---	---	----

5	14	7	4	15
---	----	---	---	----

5	7	14	4	15
---	---	----	---	----

5	7	4	14	15
---	---	---	----	----

(3)

5	7	4	14	15
---	---	---	----	----

5	4	7	14	15
---	---	---	----	----

(4)

5	4	7	14	15
---	---	---	----	----

4	5	7	14	15
---	---	---	----	----

Date

Month

Year



→ Dry Run

14 15 5 7 4

14 5 15 7 4

i j temp n

14 5 7 15 4

0 0 15 5

1

2

3

4

1 0

1

2

3

4

4 0

1

2

3

4

Insertion Sort

Algorithm:-

```

for (i = 1, i < n, i++)
{
    temp = a[i];
    j = i - 1;
    while (j > 0 && a[j] > temp)
    {
        a[j + 1] = a[j];
        j--;
    }
    a[j + 1] = temp;
}
    
```

Example:-

Sorted..

5	4	10	1	6	2
---	---	----	---	---	---

Unsorted

4	5	10	1	6	2
---	---	----	---	---	---

S

4	5	10	1	6	2
---	---	----	---	---	---

S

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DATE MONTH YEAR

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1 | 4 | 5 | 10 | 6 | 2 |
S U

1 | 4 | 5 | 6 | 10 | 2 |
S U

1 | 2 | 4 | 5 | 6 | 10 |
S

Dry Run:

5 | 4 | 10 | 1 | 6 | 2 |

i j temp n

4 | 5 | 10 | 1 | 6 | 2 |

1 0 4 6
-1

1 | 4 | 5 | 10 | 6 | 2 |

2 1 10

1 | 2 | 4 | 5 | 6 | 10 |

3 2 1
1

DATE

MONTH

YEAR



Selection Sorting

Algorithm:-

```

for (i=0; i<n-1; i++)
{
    int min=i;
    for (j=i+1; j<n; j++)
    {
        if (a[j] < a[min])
        {
            min=j;
        }
    }
    if (min != i)
    {
        temp = a[i];
        a[i] = a[min];
        a[min] = temp;
    }
}

```

Example:-

1	7	4	10	8	3	1
---	---	---	----	---	---	---

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DATE

MONTH

YEAR



1 | 4 | 10 | 8 | 3 | 7 |
S U

1 | 3 | 10 | 8 | 4 | 7 |
S U

1 | 3 | 4 | 8 | 10 | 7 |
S U

1 | 5 | 4 | 7 | 10 | 8 |
S U

1 | 5 | 4 | 7 | 8 | 10 |
S U

Dry Run:

i j min n 7 4 10 8 3 1

0 1 0 6 1 4 10 8 3 7

2 1

3 4

4 5

5

6

7

8

9